原生 Python

Astra Automation

NetApp
May 03, 2022

This PDF was generated from https://docs.netapp.com/zh-cn/astra-automation/python/python_before.html on May 03, 2022. Always check docs.netapp.com for the latest.

目录

原生 Python 1
开始之前 1
列出受管应用程序 2

原生 Python

开始之前

Python 是一种受欢迎的开发语言，尤其适用于数据中心自动化。在将 Python 的原生功能与多个通用软件包结合使用之前，您需要准备环境和所需的输入文件。

> 除了使用 Python 直接访问 Astra Control REST API 之外， NetApp 还提供了一个工具包软件包，用于抽象化 API 并消除一些复杂性。请参见 "NetApp Astra Control Python SDK" 有关详细信息 …

准备环境

下面介绍了运行 Python 脚本的基本配置要求。

Python 3.

您需要安装最新版本的 Python 3 。

其他库

必须安装 * 请求 * 和 * urllib3* 库。您可以根据环境需要使用 pip 或其他 Python 管理工具。

网络访问

运行脚本的工作站必须能够访问网络并访问 Astra Control 。使用 Astra 控制服务时，您必须连接到互联网，并且能够连接到 `https://astra.netapp.io` 上的服务。

身份信息

您需要一个具有帐户标识符和 API 令牌的有效 Astra 帐户。请参见 "获取 API 令牌" 有关详细信息 …

创建 JSON 输入文件

Python 脚本依赖于 JSON 输入文件中包含的配置信息。下面提供了示例文件。

> 您需要根据环境的具体情况更新这些示例。

身份信息

以下文件包含 API 令牌和 Astra 帐户。您需要使用 ` -i` （或 ` -identity` ） CLI 参数将此文件传递到 Python 脚本。

```
{
  "api_token": "kH4CA_uVIa8q9UuPzhJaAHaGlaR7-no901DkkrVjIXk=",
  "account_id": "5131dfdf-03a4-5218-ad4b-fe84442b9786"
}
```

列出受管应用程序

您可以使用以下脚本列出 Astra 帐户的受管应用程序。

请参见 "开始之前" 所需 JSON 输入文件的示例。

```
#!/usr/bin/env python3
##-----------------------------------------------------------------------
------
#
# Usage: python3 list_man_apps.py -i identity_file.json
#
# (C) Copyright 2021 NetApp, Inc.
#
# This sample code is provided AS IS, with no support or warranties of
# any kind, including but not limited for warranties of merchantability
# or fitness of any kind, expressed or implied. Permission to use,
# reproduce, modify and create derivatives of the sample code is granted
# solely for the purpose of researching, designing, developing and
# testing a software application product for use with NetApp products,
# provided that the above copyright notice appears in all copies and
# that the software application product is distributed pursuant to terms
# no less restrictive than those set forth herein.
#
##-----------------------------------------------------------------------
------

import argparse
import json
import requests
import urllib3
import sys

# Global variables
api_token = ""
account_id = ""

def get_managed_apps():
    ''' Get and print the list of managed apps '''
```

```
    # Global variables
    global api_token
    global account_id

    # Create an HTTP session
    sess1 = requests.Session()

    # Suppress SSL unsigned certificate warning
    urllib3.disable_warnings(urllib3.exceptions.InsecureRequestWarning)

    # Create URL
    url1 = "https://astra.netapp.io/accounts/" + account_id +
"/k8s/v1/managedApps"

    # Headers and response output
    req_headers  = {}
    resp_headers = {}
    resp_data    = {}

    # Prepare the request headers
    req_headers.clear
    req_headers['Authorization'] = "Bearer " + api_token
    req_headers['Content-Type'] = "application/astra-managedApp+json"
    req_headers['Accept'] = "application/astra-managedApp+json"

    # Make the REST call
    try:
        resp1 = sess1.request('get', url1, headers=req_headers,
allow_redirects=True, verify=False)

    except requests.exceptions.ConnectionError:
        print("Connection failed")
        sys.exit(1)

    # Retrieve the output
    http_code = resp1.status_code
    resp_headers = resp1.headers

    # Print the list of managed apps
    if resp1.ok:
        resp_data = json.loads(resp1.text)
        items = resp_data['items']
        for i in items:
            print(" ")
            print("Name: " + i['name'])
```

```
            print("ID: " + i['id'])
            print("State: " + i['state'])
    else:
        print("Failed with HTTP status code: " + str(http_code))

    print(" ")

    # Close the session
    sess1.close()

    return

def read_id_file(idf):
    ''' Read the identity file and save values '''

    # Global variables
    global api_token
    global account_id

    with open(idf) as f:
        data = json.load(f)

    api_token = data['api_token']
    account_id = data['account_id']

    return

def main(args):
    ''' Main top level function '''

    # Global variables
    global api_token
    global account_id

    # Retrieve name of JSON input file
    identity_file = args.id_file

    # Get token and account
    read_id_file(identity_file)

    # Issue REST call
    get_managed_apps()

    return

def parseArgs():
        ''' Parse the CLI input parameters '''
```

```
        parser = argparse.ArgumentParser(description='Astra REST API -
List the managed apps',
                            add_help = True)
        parser.add_argument("-i", "--identity", action="store", dest
="id_file", default=None,
                            help='(Req) Name of the identity input file',
required=True)

        return parser.parse_args()

if __name__ == '__main__':
    ''' Begin here '''

    # Parse input parameters
    args = parseArgs()

    # Call main function
    main(args)
```

Copyright Information

Copyright © 2022 NetApp, Inc. All rights reserved. Printed in the U.S. No part of this document covered by copyright may be reproduced in any form or by any means-graphic, electronic, or mechanical, including photocopying, recording, taping, or storage in an electronic retrieval system-without prior written permission of the copyright owner.

Software derived from copyrighted NetApp material is subject to the following license and disclaimer:

THIS SOFTWARE IS PROVIDED BY NETAPP “AS IS” AND WITHOUT ANY EXPRESS OR IMPLIED WARRANTIES, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE, WHICH ARE HEREBY DISCLAIMED. IN NO EVENT SHALL NETAPP BE LIABLE FOR ANY DIRECT, INDIRECT, INCIDENTAL, SPECIAL, EXEMPLARY, OR CONSEQUENTIAL DAMAGES (INCLUDING, BUT NOT LIMITED TO, PROCUREMENT OF SUBSTITUTE GOODS OR SERVICES; LOSS OF USE, DATA, OR PROFITS; OR BUSINESS INTERRUPTION) HOWEVER CAUSED AND ON ANY THEORY OF LIABILITY, WHETHER IN CONTRACT, STRICT LIABILITY, OR TORT (INCLUDING NEGLIGENCE OR OTHERWISE) ARISING IN ANY WAY OUT OF THE USE OF THIS SOFTWARE, EVEN IF ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

NetApp reserves the right to change any products described herein at any time, and without notice. NetApp assumes no responsibility or liability arising from the use of products described herein, except as expressly agreed to in writing by NetApp. The use or purchase of this product does not convey a license under any patent rights, trademark rights, or any other intellectual property rights of NetApp.

The product described in this manual may be protected by one or more U.S. patents, foreign patents, or pending applications.

RESTRICTED RIGHTS LEGEND: Use, duplication, or disclosure by the government is subject to restrictions as set forth in subparagraph (c)(1)(ii) of the Rights in Technical Data and Computer Software clause at DFARS 252.277-7103 (October 1988) and FAR 52-227-19 (June 1987).

Trademark Information

NETAPP, the NETAPP logo, and the marks listed at http://www.netapp.com/TM are trademarks of NetApp, Inc. Other company and product names may be trademarks of their respective owners.